THE BLOWUP ALGEBRA AFTER BASE CHANGE IS THE BASE CHANGE MODULO ITS TORSION BY THE ELEMENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The affine blowup algebra and its fractional description. Used to build the inverse map in the counted proof (lines 17419-17421). Lets an element be written as y/b^n with y in J^n and then lifted through the tensor product.
- Torsion by a power of the element. Used in the definition of the quotient S' at the start of the counted proof (lines 17409-17417). Kills exactly the elements that obstruct the inverse map from being well defined.

Lemma 0.1. *Let $R \to S$ be a ring map. Let $I \subset R$ be an ideal and $a \in I$. Set $J = IS$ and let $b \in J$ be the image of a. Then $S[\frac{J}{b}]$ is the quotient of $S \otimes_R R[\frac{I}{a}]$ by the ideal of elements annihilated by some power of b.*

Proof. Let S' be the quotient of $S \otimes_R R[\frac{I}{a}]$ by its b-power torsion elements. The ring map

$$S \otimes_R R[\tfrac{I}{a}] \longrightarrow S[\tfrac{J}{b}]$$

is surjective and annihilates a-power torsion as b is a nonzerodivisor in $S[\frac{J}{b}]$. Hence we obtain a surjective map $S' \to S[\frac{J}{b}]$. To see that the kernel is trivial, we construct an inverse map. Namely, let $z = y/b^n$ be an element of $S[\frac{J}{b}]$, i.e., $y \in J^n$. Write $y = \sum x_i s_i$ with $x_i \in I^n$ and $s_i \in S$. We map z to the class of $\sum s_i \otimes x_i/a^n$ in S'.

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This is well defined because an element of the kernel of the map $S \otimes_R I^n \to J^n$ is annihilated by a^n, hence maps to zero in S'. $\square$